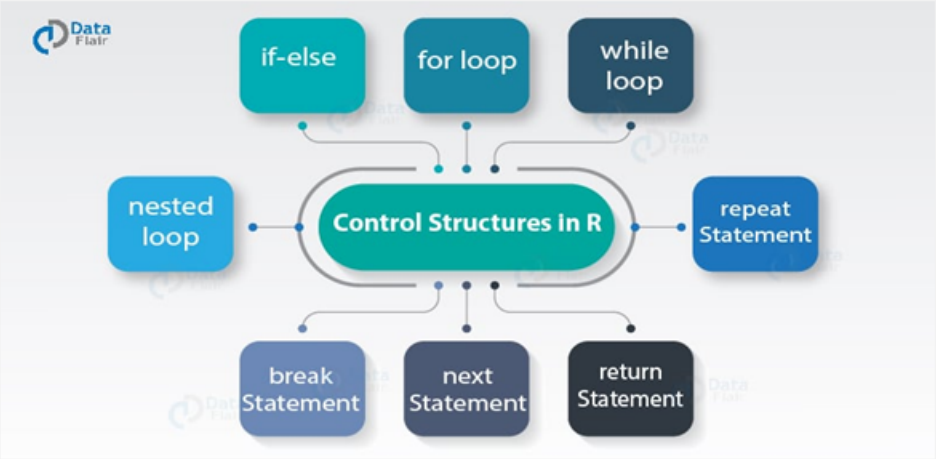
R

Control Statements

Objective

- if conditions
- switch
- for loop
- while loop
- nested loop
- repeat and break statement
- break statement
- next statement
- Summary



if condition

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Syntax:

```
if(expression)
{
    statements
    ....
    ....
}
```

```
if(expression)
{
    statements
 }else{
    statements
    }
```

```
if(condition ) {
  statement
 } else
if(condition ) {
statement
 } else {
        statement
  }
```

```
if(condition)
{
 if(condition)
   stmt
  else
   stmt
}
```

Example:

```
x <- 100
if(x > 10){
  print( x)
}
```

```
x <- 100
if(x > 10){
  print( x)
}else
{
print("small")
}
```

```
x <- 100
if(x > 10){
  print( x)
}else if(x < 10)
{
 print("small")
}else{
  print(x)
}
```

```
if(a > b)
{
  if(b>c)
    print(b)
  else
    print(c)
}
```


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switch

In this switch function expression is matched to list of cases. If a match is found then it prints that case's value. No default case is available here. If no case is matched it outputs NULL as shown in example.

Syntax:

```
Syntax:

switch (expression, case1, case2, case3,...,case n )
```

Example:

```
# Expression in terms of the index value
x <- switch(
  2,                    # Expression
  "case1", # case 1
  "case2",  # case 2
  "case3"   # case 3
)
print(x)
```


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```
a<- 12
b<- 34
choice<- readline("Enter Choice")
switch(choice,
    "add" = cat(a+b),
    "sub" = cat(a-b),
    "div" = cat(a/b),
    "pow" = cat(a^b),
    print("Invalid choice")
    )
```

```
no<-2
x<-switch(no,
        "One","Two","Three")
print(x)
```


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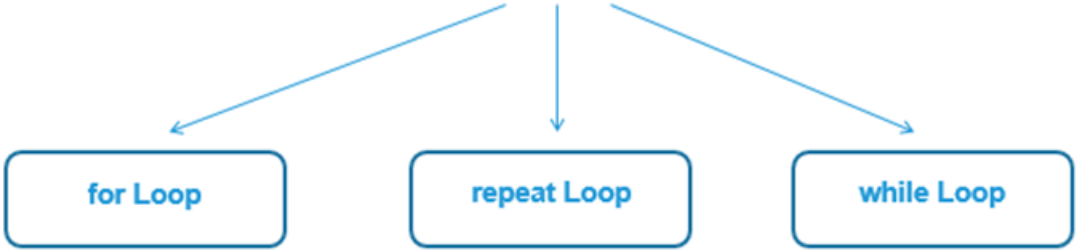
Types of Loops



- Iterates over the elements of any sequence (vector) **till the condition defined is true**
- **Number of iterations** are fixed and known in advance

- **Infinite loop** and used with **break statement** to exit the loop
- **Number of iterations** depends on the condition which is checked at the end of each iteration

- Repeats a statement or group of statements **until some condition is met**
- **Number of iterations** depends on the condition which is checked at the beginning of each iteration

Loops

- **For Loop**

```
for (variable in sequence){
   Do something
}
```

Example

```
for (i in 1:4){
   j <- i + 10
   print(j)
}
```

- **While Loop**

```
while (condition){
   Do something
}
```

Example

```
while (i < 5){
   print(i)
   i <- i + 1
}
```

```
repeat {
   statement
   statement
      ...
   statement
}
```

Loop Code will be run forever ...

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break and next statement

Break statement

break statement can be used in any type of loop to exit from the loop.

next statement

next statement is used to skip the current iteration without executing the further statements and continues the next iteration cycle without terminating the loop.

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Thank You